



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN

SEMESTER: Fall 2023-2024

I. Course Code and Title CSC 1204: Discrete Mathematics II. Credit 3 credit hours (3 hours theory per week) III. Nature Core Course for CSE IV. Prerequisite CSC 1102: Intro. to Programming Language MAT 1102: Diff. Calculus & Co-Ordinate	V. Vision: Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context. VI. Mission: The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.
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VII - Course Description:

- Explain propositional logic and propositional equivalences
- Explain different types of sets and set operations
- Determine whether two compound propositions are logically equivalent using different techniques
- Describe different types of functions, representations of Graphs
- Describe different types of Tree traversal algorithms such as Preorder, Inorder, Postorder
- Explain Euler and Hamilton paths and circuits
- Discuss Relations and their properties

VIII - Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain***			PO Assessed****
		C	P	A	
CO1**	Apply the concepts of Propositional Logic, Sets, and Functions to verify different mathematical properties	3			PO-a-1
CO2	Apply information and concepts of Mathematical Induction, Integers, Matrices, and Counting theory to solve problems	3			PO-a-2
CO3	Identify mathematical properties associated in Relation	2			PO-b-1
CO4**	Identify and formulate computing problems using Graphs and Trees	4			PO-b-2

C: Cognitive; P: Psychomotor; A: Affective Domain

* CO assessment method and rubric of COs assessment is provided in later section

** COs will be mapped with the Program Outcomes (POs) for PO attainment

*** The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.

**** The numbers under 'PO Assessed' column represent the POs each CO corresponds to.

IX - Topics to be covered in the class and/or lab: *

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1	Knowing Mission & Vision of AIUB. To present an overview of the course and discuss the course contents, Propositional Logic, Propositional Equivalences	Introduction, discussion, and student counseling	Quiz, Term Exam
Week 2	CO1	Predicates and Quantifiers, Sets, Set Operations, bit string as a set operation	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 3	CO1	Functions, different types of functions – One-to-one, Onto, One-to-one correspondence, Inverse function, Composite function	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 4	CO1	The Integers and Division, Fundamental Theorem of Arithmetic, Division Algorithm, Primes and Greatest Common Divisors, Least Common Multiples, Primality, Relative Primality, Modular arithmetic, Congruence	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 5	CO2	Caesar's Cipher, Matrices, Matrix arithmetic and Boolean operations, Zero-one matrix	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 6	CO1, CO2	Methods of Proof, Proof by Mathematical Induction	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 7		Revision		
Midterm (Week 8)				
Week 9	CO2	The Basics of Counting, Sum Rule, Product Rule, The Pigeonhole Principle	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 10	CO3	Relations and Their Properties, Representing Relations using matrix and directed Graph	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 11	CO4	Introduction to Graphs, Graph Terminology Representing Graphs and Graph Isomorphism, Connectivity	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 12	CO4	Euler and Hamilton Paths & circuits	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 13	CO4	Planar Graphs, Graph Coloring, Introduction to Trees, Applications of Trees, Binary Search Tree	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 14	CO4	Tree Traversal, Infix, Prefix and Postfix Notations	Lecture slide, board work, problem solving, book discussion and group study	Quiz, Term Exam
Week 15		Revision		
Final term (Week 16)				

* The faculty reserves the right to change, amend, add, or delete any of the contents.

X - Mapping of PO to Courses and K, P, A

PO Indicator ID	PO Indicators Definition (As per the requirement of Wks)	Domain	K	P	A
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PO-a-1	Apply information and concepts in natural science with the familiarity of issues.	Cognitive Level 3 (Applying)	K1		
PO-a-2	Apply information and concepts of mathematics with the familiarity of issues.	Cognitive Level 3 (Applying)	K2		
PO-b-1	Identify first principles of natural sciences and engineering sciences in practical applications.	Cognitive Level 2 (Understanding)	K1		
PO-b-2	Formulate solutions, procedures, and methods using first principles of mathematics for engineering sciences.	Cognitive Level 4 (Analyzing)	K2		

XI – K, P, A Definitions

Indicator	Title	Description
K1	Theory based natural science	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Conceptual based mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	Apply the concepts of Propositional Logic, Sets, and Functions to verify different mathematical properties	PO-a-1	Quiz/ Term Exam	Rubric for Term Exam
CO2	Apply information and concepts of Mathematical Induction, Integers, Matrices, and Counting theory to solve problems	PO-a-2	Quiz/ Term Exam	Rubric for Quiz/ Term Exam
CO3	Identify mathematical properties associated in Relation	PO-b-1	Quiz/ Term Exam	Rubric for Quiz/ Term Exam
CO4	Identify and formulate computing problems using Graphs and Trees	PO-b-2	Quiz/ Term Exam	Rubric for Term Exam

XIII – Evaluation and Assessment Criteria

CO1: Apply the concepts of Propositional Logic, Sets, and Functions to verify different mathematical properties					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Knowledge of	Apply information and concepts in natural science for relevant mathematical problems				

Natural Science	
Mathematical Knowledge	Understanding the mathematical terminologies and their application in the Computer Science arena
Analysis of the problems or issues	Comprehensive and in-depth analysis of problems or issues
Understanding	Understanding the given problem with the conception from the topics.

CO2: Apply information and concepts of Mathematical Induction, Integers, Matrices, and Counting theory to solve problems

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Completeness	Understanding the appropriate procedure to complete the mathematical process				
Mathematical Knowledge	Understanding the mathematical terminologies and their application in the Computer Science arena.				
Content knowledge	Present accurate and full knowledge of the applications of mathematical problems				

CO3: Identify mathematical properties associated with Relation

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Problem Analysis	Explain the real concern in the context of relations and why it is crucial to investigate it				
Mathematical Knowledge	Explore the mathematical terminologies and their usage in the Computer Science arena				
Completeness	Understanding the appropriate methods to solve problems or issues of the Relation				

CO4: Identify and formulate computing problems using Graphs and Trees

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Fundamental concepts and principles	Demonstrate a clear understanding and concepts of the theory and the applications of Graphs and Trees				
Problem Analysis	Clearly identifies and summarizes the concerns, issues, and problems related to Graphs and Trees				

XIV- Course Requirements

- Students are expected to attend at least 80% class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignment and projects should be in due time.

XV – Evaluation & Grading System

The following grading system will be strictly followed in this class

MID TERM		FINAL TERM	
Attendance	10%	Attendance	10%
Quiz	40%	Quiz	40%
Mid-Term written exam	50%	Final Term written exam	50%
Total	100%	Total	100%
Grand Total 100% = 40% of Midterm + 60% of Final Term			

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

* The evaluation system will be strictly followed as par the AIUB grading policy.

XVI – Textbook/ References

- DISCRETE MATHEMATICS AND ITS APPLICATIONS WITH COMBINATORICS AND GRAPH THEORY (7TH EDITION) BY KENNETH H. ROSEN [INDIAN ADAPTATION BY KAMALA KRITHIVASAN], PUBLISHED BY MCGRAW-HILL
- DISCRETE MATHEMATICS, RICHARD JOHNSONBAUGH, PEARSON EDUCATION, INC.
- DISCRETE MATHEMATICAL STRUCTURES, BERNARD KOLMAN, ROBERT C. BUSBY, SHARON ROSS, PRENTICE-HALL, INC.
- DISCRETE AND COMBINATORIAL MATHEMATICS, R. P. GRIMALDI, 3RD EDITION, ADDISON-WESLEY
- SCHAUM'S OUTLINES DISCRETE MATHEMATICS (2ND EDITION), BY SEYMOUR LIPSCHUTZ, MARC LIPSON
- DISCRETE MATHEMATICS MODELS, F. S. ROBERTS, PRENTICE HALL
- LECTURE SLIDES

XVII - List of Faculties Teaching the Course

FACULTY NAME	SIGNATURE
AFFSAH SHARMIN	
DR. ABDUS SALAM	
DR. ASHRAF UDDIN (Convenor)	

DR. M. MAHMUDUL HASAN	
DR. MD. IFTEKHARUL MOBIN	
DR. MD. MEHEDI HASAN	
DR. MOHAMMAD RABIUL ISLAM	
DR. S M HASAN MAHMUD	
FARZANA BENTE ALAM	
ISRAT JAHAN MOURI	
MD. HASIBUL HASAN	
MD. SAJID BIN FAISAL	
NYME AHMED	
SAEEDA SHARMEEN RAHMAN	
SHARIFA RANIA MAHMUD	
SUPTA RICHARD PHILIP	
UMME SADIA SALSABIL	
VICTOR STANY ROZARIO	
PRITAM KHAN BONI	
MD. HASIBUL ISLAM (ICE)	
QUAZI FARIHA TASNIM (ICE)	
SAMIUR RAHMAN KHAN (ICE)	
DR. MOUSHUMI ZAMAN BONNY	
SIRAJUM MUNIRA SHIFAT	

XVI – Verification

Prepared by: ----- Dr. Ashraf Uddin <i>Course Convener</i> Date:.....	Moderated by: ----- Dr. M. Mahmudul Hasan <i>Point of Contact</i> <i>OBE Implementation Committee for CS</i> Date:.....	
Checked by: ----- Dr. Akinul Islam Jony <i>Head (Undergraduate Program),</i> <i>Department of Computer Science</i> Date:.....	Certified by: ----- Dr. Md. Abdullah-Al-Jubair <i>Director,</i> <i>Faculty of Science & Technology</i> Date:.....	Approved by: ----- Dr. Dip Nandi <i>Associate Dean,</i> <i>Faculty of Science & Technology</i> Date:.....